THEJOVATI NARAYANAN

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Data Science & Analytics

EDUCATION

B.Tech + MBA 5 yr (Integrated) - Data Science 2024 – 2029

SVKM's NMIMS MPSTME, Mumbai – Completed Year 2, Semester 4, Currently in Year3, Sem5 | CGPA: 7.91

- Year 1-2 coursework: Data Structures & Algorithms, Statistical Structures & Inference, Stochastic Processes & Applications, Data Wrangling, DBMS, Optimization Methods, Data Handling & Visualization, Introduction to Data, Signal & Image Analysis, Mobile Application Development, Managing Uncertainty etc.
- Year 3 : Applied Artificial Intelligence, Foundations of Machine Learning, Computer Vision & Applications, Advanced Data Analytics, Big Data Analytics, Cloud Computing, Data Driven Program Analysis etc.
- SAT: Qualified (May 2024)

TECHNICAL SKILLS

- Languages & Tools:
  - Python (NumPy, Pandas, Scikit-learn, XGBoost, FastAPI, SHAP), SQL / MySQL, R, AWS
- Data & Analytics:
  - Tableau, Power BI, Excel, Jupyter, VS Code, Docker
- Other:
  - n8n (workflow automation), Supabase, Android Studio, AutoCAD

PROJECTS

VitalEdge+ - AI Clinical Intelligence System for Allergy Risk Detection | *Python · XGBoost · FastAPI · SHAP*

- Built an end-to-end AI system for real-time medication–allergy conflict detection using a dual-engine approach (rule-based + XGBoost), achieving near-perfect accuracy.
- Introduced an Allergy Burden Score and integrated SHAP explainability for transparent, auditable predictions in high-stakes clinical settings.
- Deployed via FastAPI with a real-time React dashboard for point-of-care decision support across clinical workflows.

Healthcare Data Warehouse Pipeline | *Python · ETL · SCD Type 2 · Star Schema · SQL*

- Designed and implemented a full ETL pipeline to consolidate siloed patient records, lab results, and prescriptions from disparate healthcare systems into a unified analytical warehouse.
- Implemented SCD Type 2 for full historical tracking and modelled a Star Schema for efficient, analytics-ready querying and reporting.

Influencer Analytics & Brand Collaboration Platform | *PostgreSQL · FastAPI · Python · Chart.js · DBeaver*

- Designed and built a full-stack analytics platform from scratch: 8-table PostgreSQL schema (3NF, ~20,000 rows across influencers, brands, campaigns, collaborations, posts) and FastAPI backend (JWT auth, 25+ endpoints, analytics APIs).
- Developed a live dashboard with KPI visualizations, filters, and real-time updates for campaign performance and engagement insights, all sourcing data from authenticated API calls.

AWS Serverless Credit Card Fraud Detection System | *Python · AWS Lambda · API Gateway · DynamoDB · Terraform*

- Designed and deployed a fully serverless fraud detection pipeline (Lambda + API Gateway + DynamoDB, zero EC2/ALB/VPC, ~25 AWS resources via Terraform, effectively $0 cost) on the ULB dataset (284,807 transactions, 0.173% fraud rate), reimplementing inference as a zero-dependency, pure-Python Lambda handler (~15KB, no ML libraries) verified byte-identical to the original scikit-learn model.
- Tuned a Logistic Regression model's decision threshold on a held-out validation set to correct for severe class imbalance, achieving 0.89 precision / 0.79 recall / 0.83 F1 and 0.966 ROC-AUC on an untouched test set, and validated the live deployed API against 50 held-out transactions (45/50 correct, 0 false positives).

AI Startup Boardroom - Multi-Agent Investment Committee System | *Python · FastAPI · React · LLM Orchestration*

- Built a 7-agent system (CFO, Market, Founder, Traction, Risk, Investor, CEO) that independently analyzes a startup from a 100,000-row dataset and deliberates through a structured 4-round protocol independent analysis → cross-examination → bounded revision → final decision , with a disagreement-detection engine that pairs the most bullish and most skeptical agents from structured position data (not LLM sentiment) and auto-detects stalemates to bound debate length.
- Built a deterministic, dataset-grounded Mock Mode for zero-cost demoing alongside a swappable real LLM mode, validated by 47 automated tests covering agents, orchestration, and the API layer.

INTERNSHIP (Wisdomleaf Technologies Pvt. Ltd.)

Jun 2026 –Present

- Developed an AI-powered Prescription Parser assigned by the company, contributing to OCR integration, image preprocessing, data extraction, backend API development, medicine validation, and testing, leading to the successful completion of Phase 1 of the project.
- Based on the Phase 1 outcome, the company offered a paid extension on the internship (Phase 2). Developed a parser for digital prescriptions as part of Phase 2, enabling structured extraction of prescription data from digital prescription formats.
- Currently developing a handwritten prescription parser without using AI, focusing on image processing, text extraction, and reliable parsing of handwritten prescription data.

LEADERSHIP & EXTRACURRICULARS

Head of Corporate Relations & Marketing 2024 – 2025

College Committee, NMIMS MPSTME

- Promoted from Executive to Head within one year, overseeing industry outreach and committee-level brand initiatives.
- Coordinated with industry leaders and senior managers across sectors, managing end-to-end corporate engagement for college events.
- Led marketing strategy and communications for committee-level campaigns, driving awareness and participation across the student body.

Participated in SIH (Smart India Hackathon) 2025 and 2026

COMMUNITY SERVICE

Volunteer — Anandam Trust Chennai, 3 Weeks (2025–26)

Anandam Free Home for Senior Citizens & Anandam Special School

- Delivered digital literacy sessions to children with autism
- Taught MS office and internet basics to underprivileged women
- Taught across three ability groups at the Special School and facilitated numeracy, language, and creative activities
- Catalogued the library with section-wise registered records for elder accessibility

RESEARCH

- "A Hybrid LLM Framework for Emission-Constrained Supply Chains with Renewable-Aware RL" – Paper accepted in COMSYS-2026, Developed a closed-loop model using LSTM/GRU, NSGA-II, PPO, RL, Monte Carlo, and ABM (MAPE 7.4%) that cut CO_2 40.3% (9.97→5.95 kg) with zero constraint violations across 500 stress scenarios, with PPO adding a further −12% emissions / +8% reward over NSGA-II alone.
- "Quantum Key Distribution using Polarization" - Research synthesis paper presented at Polarizer'25, placed second.